

**RAFFLES INSTITUTION
2018 YEAR 6 PRELIMINARY EXAMINATION**

Higher 1



CHEMISTRY

Paper 1 Multiple Choice

8873/01

24 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Do not open this question booklet until you are told to do so.

Write in **soft pencil**.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number in the spaces provided on the Answer Sheet.

There are **thirty** questions in this section. Answer **all** questions.

For each question there are four possible answers **A, B, C** and **D**.

Choose the one you consider correct and record your choice in soft pencil on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

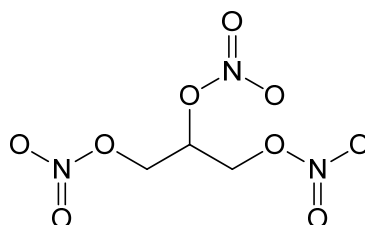
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in the question booklet.

This document consists of **15** printed pages.

For each question, there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 Nitroglycerin is an explosive that decomposes to form carbon dioxide, water, nitrogen and oxygen only.



nitroglycerin

Which row shows the number of moles of the products formed when 4 mol of nitroglycerin decomposes?

	carbon dioxide	water	nitrogen	oxygen
A	8	8	6	6
B	8	8	12	12
C	12	10	6	1
D	12	10	12	2

- 2 *Use of the Data Booklet is relevant to this question.*

A 10.0 g sample of H_2S was collected. How many H atoms are in this sample of H_2S ?
[L = Avogadro constant]

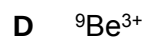
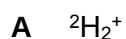
A $\frac{10.0 \times L \times 2}{34.1}$

C $\frac{2L}{10.0 \times 34.1}$

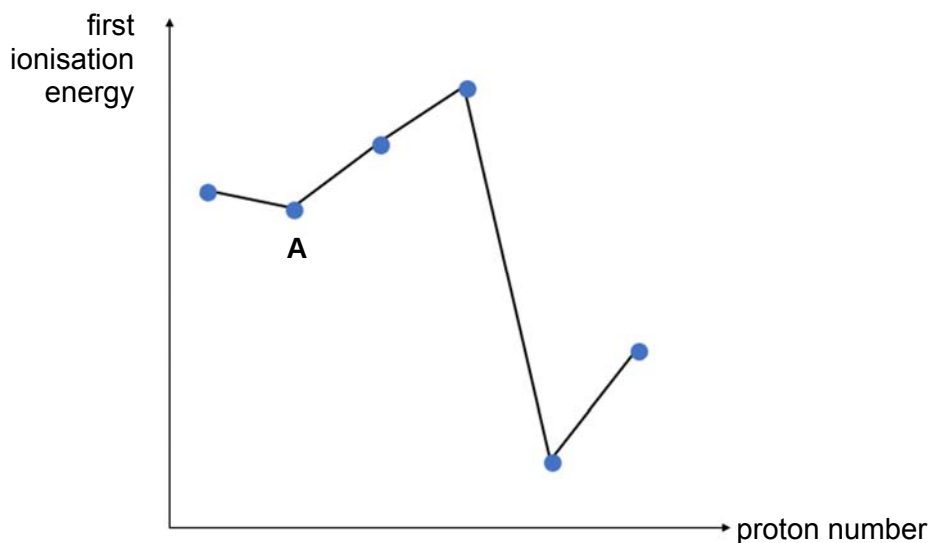
B $\frac{L \times 34.1}{2 \times 10.0}$

D $\frac{2 \times 10.0 \times 34.1}{L}$

- 3 Which of the following species would undergo the greatest deflection in an electric field?



- 4 The diagram shows the variation of first ionisation energy with proton number for seven elements of consecutive proton numbers between 1 and 20.

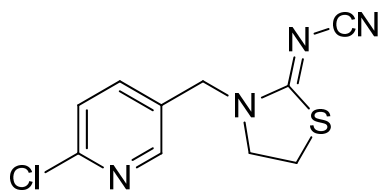


Which of the following is element **A**?

- | | |
|-------------|------------|
| A Mg | C S |
| B Si | D K |
- 5 The boiling point of hexane is 69 °C while the boiling point of 2-methylpentane is 60 °C. Which statement best explains the higher boiling point of hexane?

- A** Hexane has a higher density as it has better packing.
- B** Hexane has stronger intermolecular permanent dipole-permanent dipole interactions as it is more polar.
- C** Hexane has stronger intermolecular instantaneous dipole-induced dipole interactions as it has a larger surface area.
- D** Hexane has stronger intermolecular instantaneous dipole-induced dipole interactions as it has a larger number of electrons.

- 6 Thiacloprid is an insecticide widely used for pest control in agriculture.



thiacloprid

Which of the following shows the correct number of σ and π bonds in thiacloprid?

	σ bonds	π bonds
A	16	4
B	16	6
C	26	4
D	26	6

- 7 Which of the following statements can be explained by the presence of intermolecular hydrogen bonding?

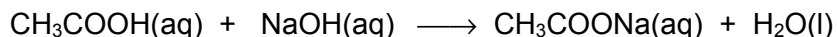
- 1 HF is a weak acid.
- 2 NaCl readily dissolves in water.
- 3 Ice has a lower density than water.
- 4 The relative molecular mass of ethanoic acid in benzene is 120.

A	1, 2 and 3	C	2 and 3
B	1 and 4	D	3 and 4

- 8 Which of the following shows the correct number of bonded electron pairs, number of lone pairs and the shape around each N atom in N_2H_4 ?

	number of bonded electron pairs around N atom	number of lone pairs around N atom	shape
A	3	0	trigonal planar
B	3	1	trigonal pyramidal
C	3	1	tetrahedral
D	4	0	tetrahedral

- 9 When 25.0 cm³ of 0.70 mol dm⁻³ NaOH(aq) is added to 25.0 cm³ of 0.70 mol dm⁻³ CH₃COOH(aq), the temperature of the mixture rise by 4.7 °C.

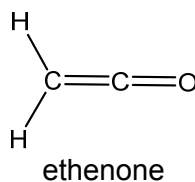


Assume that the density and specific heat capacity of the final mixture are 1.0 g cm⁻³ and 4.18 J g⁻¹ K⁻¹ respectively.

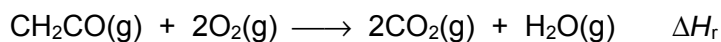
What is the enthalpy change of neutralisation for the above process?

- A +28.1 kJ mol⁻¹
 B +56.1 kJ mol⁻¹
 C -28.1 kJ mol⁻¹
 D -56.1 kJ mol⁻¹
- 10 Use of the Data Booklet is relevant to this question.

Ethenone, CH₂CO, has the following structure.



The reaction of gaseous ethenone with oxygen gas is given below.



What is the enthalpy change, ΔH_r , for this reaction?

- A -632 kJ mol⁻¹
 B -718 kJ mol⁻¹
 C -762 kJ mol⁻¹
 D -978 kJ mol⁻¹

11 Which statement about the order of a chemical reaction is correct?

- A It is the sum of the number of species included in the rate equation.
- B It is the sum of the number of species involved before the rate determining step.
- C It is the sum of the powers of the concentrations of species included in the rate equation.
- D It is the sum of the number of moles on the left-hand side of the balanced chemical equation.

12 In a kinetic study of the reaction $D + E \rightarrow \text{products}$ at constant temperature, the following data were obtained.

[D] / mol dm ⁻³	[E] / mol dm ⁻³	initial rate / mol dm ⁻³ s ⁻¹
0.01	0.05	1.9×10^{-4}
0.02	0.05	3.9×10^{-4}
0.02	0.01	4.0×10^{-4}

If the rate constant doubles for each 10 °C rise in temperature, which of the following will give the greatest rate of reaction?

	[D] / mol dm ⁻³	[E] / mol dm ⁻³	temperature / °C
A	0.1	0.2	40
B	0.2	0.2	30
C	0.3	0.4	20
D	0.4	0.1	30

13 The initial rate of the reaction between aqueous H₂O₂ and acidified KI solution was found to be first order with respect to both H₂O₂ and I⁻, and zero order with respect to H⁺.

What is the units for the rate constant of this reaction?

- A s⁻¹
- B mol dm⁻³ s
- C mol⁻¹ dm³ s⁻¹
- D mol⁻² dm⁶ s⁻¹

14 Which of the following is correct for a system at dynamic equilibrium?

- A The rates of both forward and backward reactions are zero.
- B The concentration of reactants is equal to the concentration of products.
- C There is no net change in the concentrations of the reactants and products.
- D The rate constant of the forward reaction is equal to the rate constant of the backward reaction.

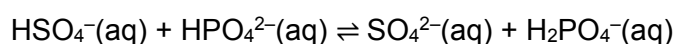
15 An equilibrium is represented by the following equation.



Which of the following is **incorrect** for the indicated changes?

	indicated changes	position of equilibrium	forward rate	backward rate
A	increase total pressure by compression	no change	increase	increase
B	increase temperature	shifts left	decrease	increase
C	adding catalyst	no change	increase	increase
D	adding more F	shifts right	increase	no change

16 Which of the following is a conjugate acid-base pair in the reaction below?



	base	conjugate acid
A	HPO_4^{2-}	H_2PO_4^-
B	HPO_4^{2-}	HSO_4^-
C	HSO_4^-	HPO_4^{2-}
D	HSO_4^-	SO_4^{2-}

- 17 The ionic product of water, K_w , at two different temperatures are given below.

temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$
25	1.00×10^{-14}
62	1.00×10^{-13}

Which statements are **incorrect** for pure water?

- 1 The self-ionisation of water is an exothermic process.
- 2 At 62 °C, $\text{pH} = 14 - \text{pOH}$.
- 3 At 62 °C, $\text{pH} < \text{pOH}$.

- | | |
|-----------------|-----------------------|
| A 1 only | C 2 and 3 only |
| B 3 only | D 1, 2 and 3 |

- 18 The first ionisation energies, in kJ mol^{-1} , of a sequence of elements of increasing proton (atomic number) are given below.

519 900 799 1090 1400 1310

Where is this sequence of elements likely to be located in the Periodic Table?

- A** Group 2
- B** Group 17
- C** from Li to N
- D** from N to Na

- 19 **L** and **M** are elements in the Periodic Table.

Element **L** forms an insoluble oxide.

Element **M** forms an acidic oxide that is unable to react with dilute NaOH.

What is the identity of elements **L** and **M**?

	L	M
A	Na	Si
B	Mg	S
C	Al	Si
D	Si	P

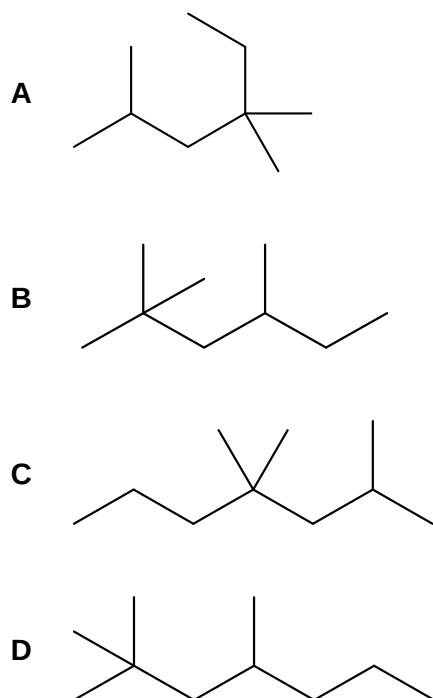
- 20 The table shows the results of experiments in which the halogens X_2 , Y_2 and Z_2 were added to separate aqueous solutions containing X^- , Y^- and Z^- ions.

	$X^-(aq)$	$Y^-(aq)$	$Z^-(aq)$
X_2	no reaction	no reaction	no reaction
Y_2	X_2 formed	no reaction	Z_2 formed
Z_2	X_2 formed	no reaction	no reaction

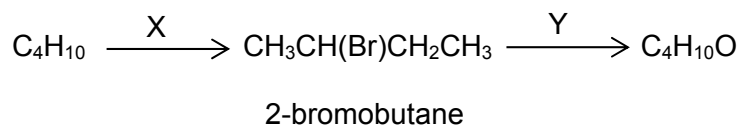
Which of the following shows X_2 , Y_2 and Z_2 in order of increasing oxidising strength?

- A $X_2 \rightarrow Y_2 \rightarrow Z_2$
 B $X_2 \rightarrow Z_2 \rightarrow Y_2$
 C $Y_2 \rightarrow Z_2 \rightarrow X_2$
 D $Y_2 \rightarrow X_2 \rightarrow Z_2$

- 21 What is the skeletal formula of 2,4,4-trimethylhexane?



- 22 The diagram shows a reaction scheme involving 2-bromobutane.



What types of reaction are reaction X and reaction Y?

	X	Y
A	addition	substitution
B	substitution	substitution
C	addition	oxidation
D	substitution	oxidation

- 23 How many different alkenes, including *cis-trans* isomers, could be produced when $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)\text{BrCH}_2\text{CH}_2\text{CH}_3$ is treated with hot ethanolic KOH?

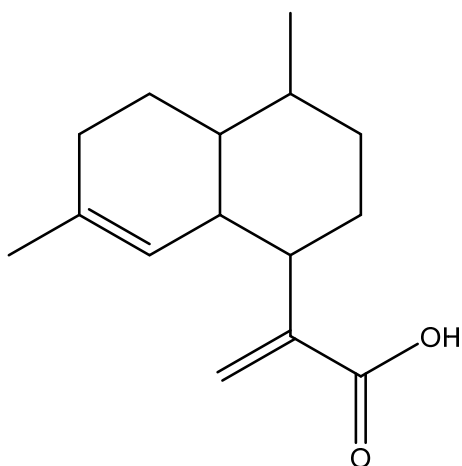
- | | |
|-----|-----|
| A 2 | C 4 |
| B 3 | D 5 |

- 24 An ester, W, of molecular formula $\text{C}_6\text{H}_{12}\text{O}_2$ undergoes acid hydrolysis to produce a carboxylic acid and an alcohol. The alcohol is found to have no reaction with hot acidified sodium dichromate(VI).

What could be the structural formula of ester W?

- A $\text{CH}_3\text{CO}_2\text{C}(\text{CH}_3)_3$
 B $(\text{CH}_3)_3\text{CCO}_2\text{CH}_3$
 C $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}(\text{CH}_3)_2$
 D $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}_3$

- 25 Artemisinic acid is a useful intermediate for making the anti-malarial drug, artemisin.



artemisinic acid

Which statements about artemisinic acid are correct?

- 1 It decolourises two moles of greenish-yellow chlorine.
- 2 It can be formed from the oxidation of an aldehyde.
- 3 It can be reduced by hydrogen gas.

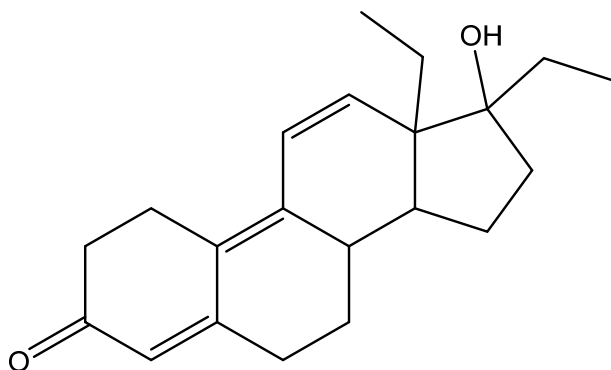
A 1,2 and 3

C 1 and 3

B 1 and 2

D 2 and 3

- 26** THG is an anabolic steroid that mimics the effects of natural hormones, like testosterone, found in the body. It promotes muscle growth and its misuse has led to many athletes being disqualified from international competitions, such as The Olympic Games.



THG

Which statements about THG are correct?

- 1 It forms a C=C bond when heated with dilute sulfuric acid.
- 2 It reacts with sodium borohydride to form a secondary alcohol.
- 3 It is oxidised by potassium dichromate(VI) to a carboxylic acid.

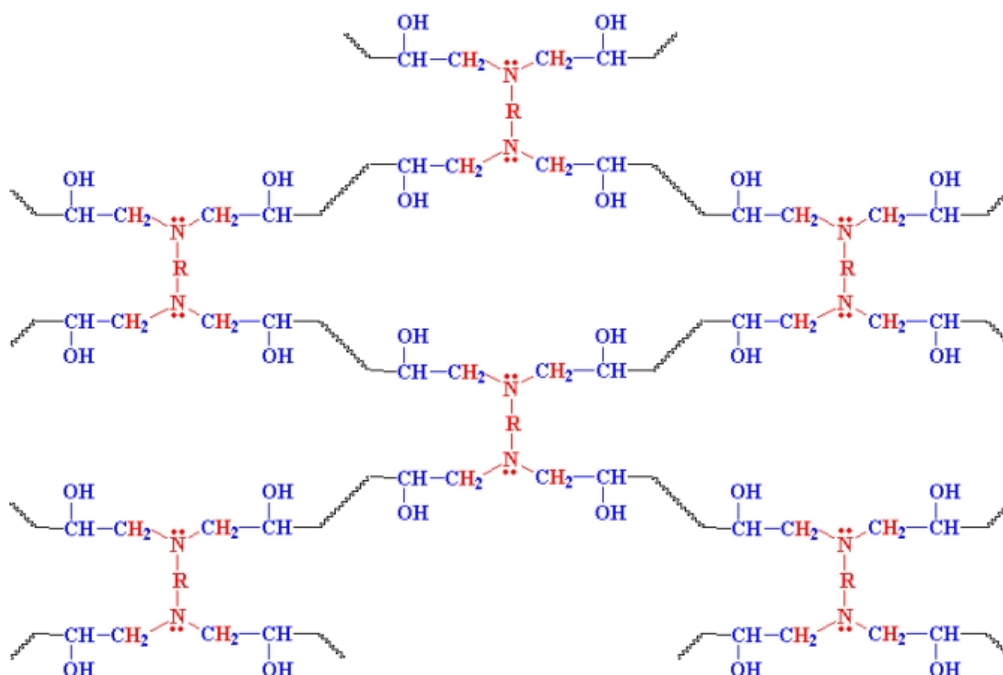
A 2 only

C 1 and 3

B 1 and 2

D 2 and 3

- 27 Epoxy resins are renowned for their strong adhesive qualities. A segment of an epoxy resin is given below.



epoxy resin

Which statements about an epoxy resin are correct?

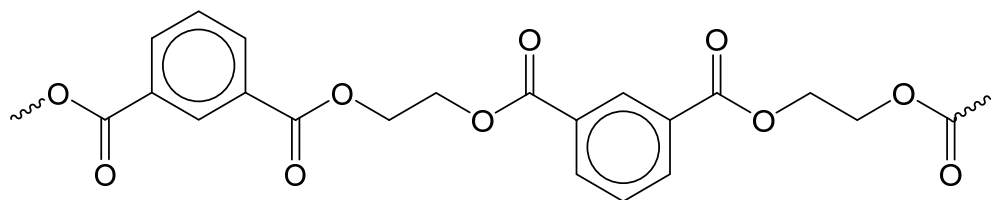
- 1 It is a thermoset.
- 2 It contains the amide functional group.
- 3 It forms strong polar bonds with hydroxyl surfaces.

- A** 1 and 2
B 1 and 3

- C** 2 and 3
D 1, 2 and 3

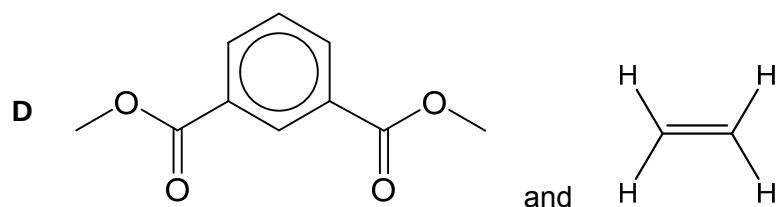
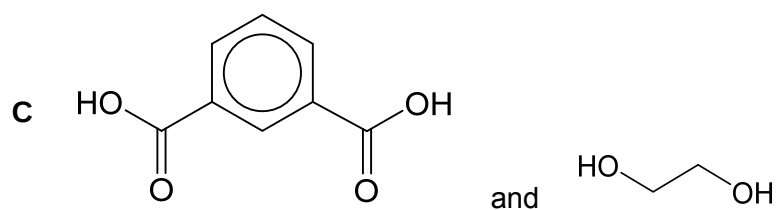
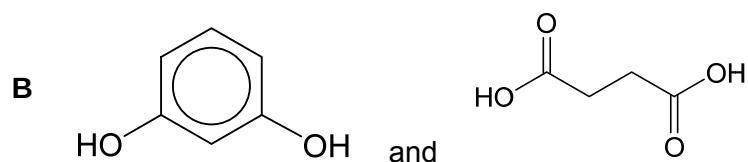
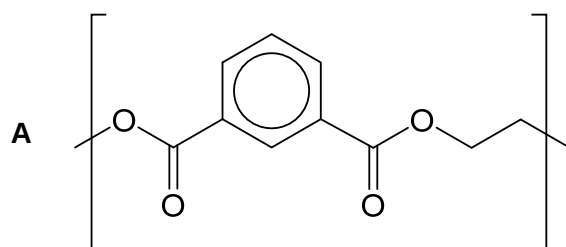
- 28 Football jerseys are commonly made from polymers, such as PET, as it is lightweight, durable and only absorbs 0.4% of its weight of water.

A polymer chain of PET is given below.



PET

Which are the monomers of PET?



29 What is the maximum size, in at least one dimension, of a nanomaterial?

- A 1×10^{-6} m
- B 1×10^{-7} m
- C 1×10^{-8} m
- D 1×10^{-9} m

30 The honeycomb structure is commonly used to make blankets and is also found in catalytic converters and graphene.

Which statement about the honeycomb structure is **incorrect**?

- A It allows air to be trapped.
- B It will be easily deformed by pressure.
- C Its large surface area to volume ratio increases catalytic efficiency.
- D It has high tensile strength due to the network of strong covalent bonds within each layer.

– END OF PAPER –